Elimination and Back Substitution

(Linear Algebra)

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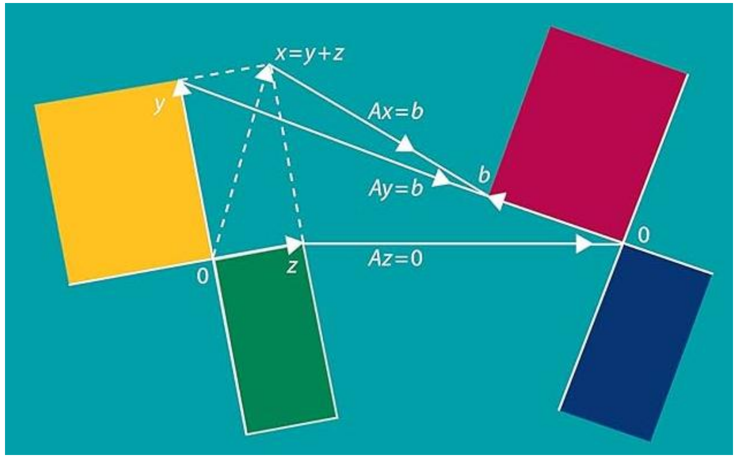


Matrix Multiplication

1 To multiply AB we need *row length for* A = *column length for* B.

2 The number in row i, column j of AB is **(row $\boldsymbol{i}$ of $\boldsymbol{A}$) · (column $\boldsymbol{j}$ of $\boldsymbol{B}$)**.

3 By columns: **$\boldsymbol{A}$ times column $\boldsymbol{j}$ of $\boldsymbol{B}$ produces column $\boldsymbol{j}$ of $\boldsymbol{AB}$.**

4 Usually AB is different from BA. But always $(\boldsymbol{AB})\boldsymbol{C} = \boldsymbol{A}(\boldsymbol{BC})$.

5 If A has r independent columns in C, then $\boldsymbol{A} = \boldsymbol{CR} = (\boldsymbol{m} \times \boldsymbol{r})(\boldsymbol{r} \times \boldsymbol{n})$.

$$\boxed{\begin{array}{c}\textbf{Column } j \textbf{ of } AB \textbf{ equals } A \textbf{ times column } j \textbf{ of } B \\[2ex] \textbf{If } B = \begin{bmatrix} \\ b_1 \cdots b_p \\ \\ \end{bmatrix} \textbf{ then } AB = \begin{bmatrix} \\ Ab_1 \cdots Ab_p \\ \\ \end{bmatrix}\end{array}}$$

$$Ab_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$Ab_2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

Row way Dot Product

$$Ab_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix} = \begin{bmatrix} \text{row } 1 \cdot b_1 \\ \text{row } 2 \cdot b_1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 5 + 2 \cdot 7 \\ 3 \cdot 5 + 4 \cdot 7 \end{bmatrix} = \begin{bmatrix} 19 \\ 43 \end{bmatrix}$$

Column way Dot Product

$$Ab_1 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 5 \\ 7 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 7 \begin{bmatrix} 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 15 \end{bmatrix} + \begin{bmatrix} 14 \\ 28 \end{bmatrix} = \begin{bmatrix} 19 \\ 43 \end{bmatrix}$$

AB and BA

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Multiply AB and BA and check whether those two multiplications give the same answer.

AB and BA

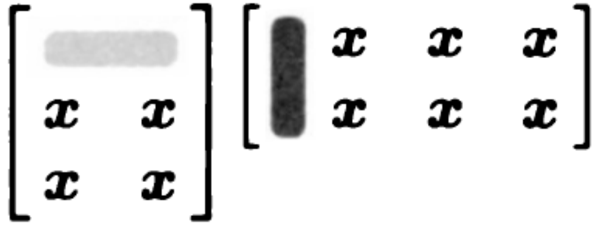
$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$AB = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} \qquad BA = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}$$

Matrix multiplication is not commutative.

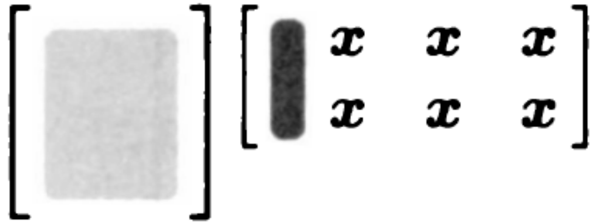
$$BA \neq AB$$

(3 by 2) (2 by 4) = (3 by 4) Four Ways to Multiply $AB = C$



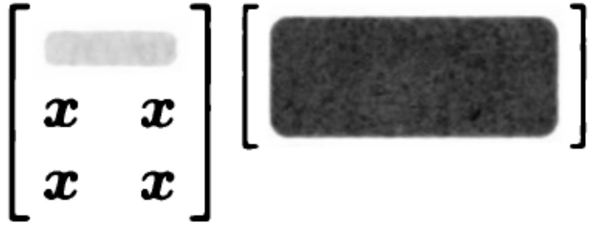
(**Row** i of A) · (**Column** k of B) = **Number** C_{ik}

$i = 1$ to 3 $\quad k = 1$ to 4 $\quad$ **12 numbers**



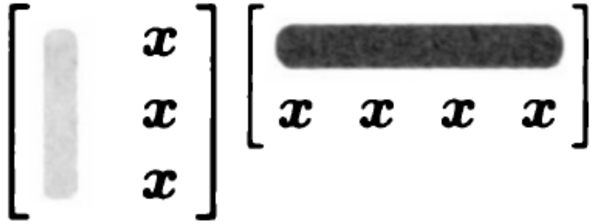
A times (**Column** k of B) = **Column** k **of** C

$k = 1$ to 4 $\quad$ **4 columns**



(**Row** i of A) times B = **Row** i **of** C

$i = 1$ to 3 $\quad$ **3 rows**



(**Column** j of A) (**Row** j of B) = **Rank 1 Matrix**

$j = 1$ to 2 $\quad$ **2 matrices**

Questions -1

Elimination

- The elimination method is a technique used to solve systems of linear equations.
- Methods:
  - Eliminating one variable
  - Creating a simpler equation with one variable
  - Solving step by step

$$x + y = 5$$

$$x - y = 1$$

Gaussian Elimination

- **Gaussian elimination** is a systematic method in Linear Algebra used to solve systems of linear equations by converting the system into an **upper triangular form** using row operations, and then solving by **back substitution**.

- Advantages:
  - Works for any number of equations
  - Converts system into triangular form
  - Efficient and widely used in computation

Gaussian Elimination

- We consider n x n matrix.
- $Ax = b$ gives n equations and those equations have n unknowns.
- Often but not always there is one solution x for each b.

A has an inverse A^{-1} with $A^{-1}A = I$ and $AA^{-1} = I$

Multiplying $Ax = b$ by A^{-1} produces the symbolic solution $x = A^{-1}b$.

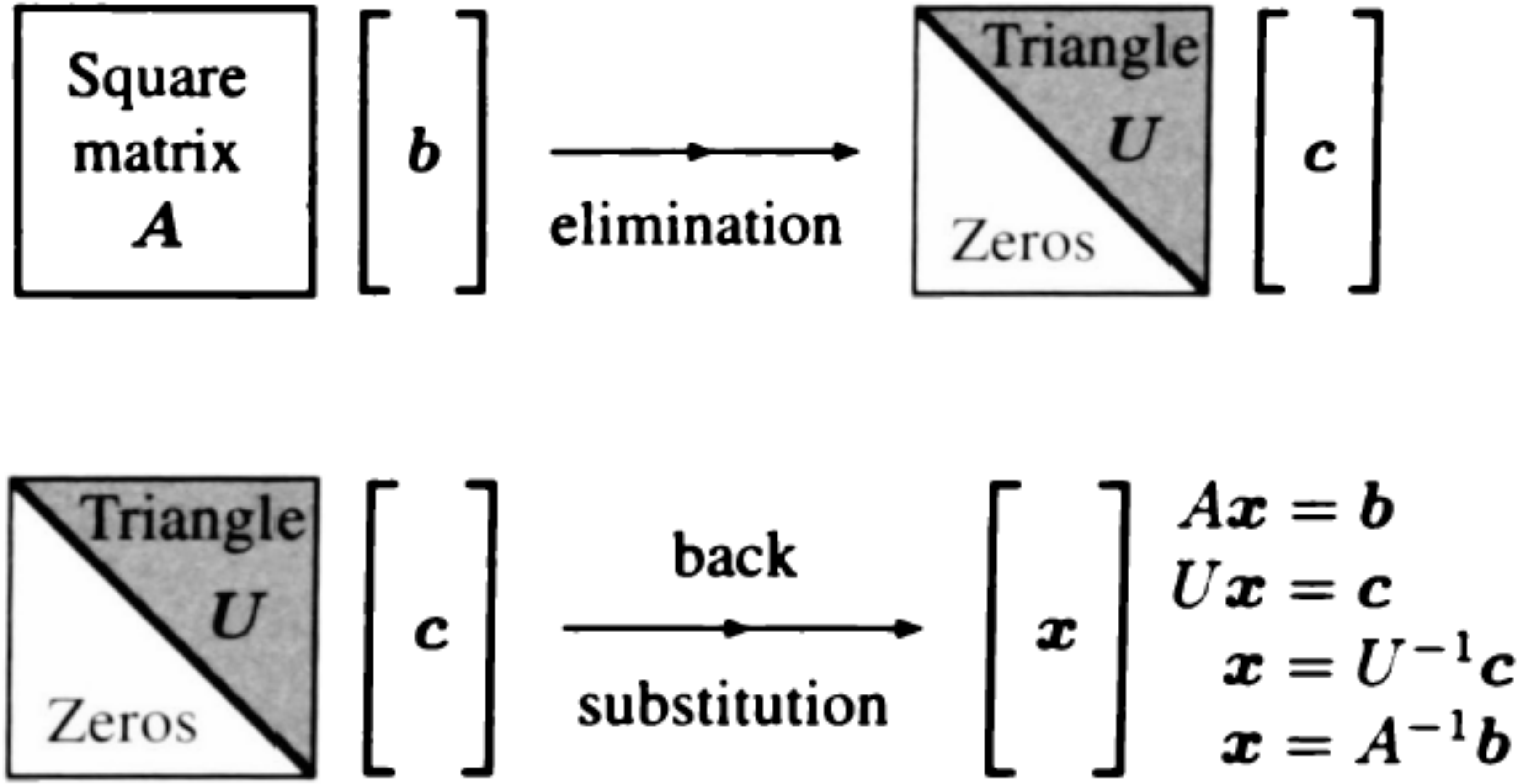
Do not compute A^{-1}

To describe all the steps we need matrices. This is the point of linear algebra! A simple elimination matrix E_{ij} produces a zero where row i meets column j ($i > j$). Overall, an elimination matrix E multiplies A to give $EA = U$. And we multiply U by an **inverse matrix** $L = E^{-1}$ to come back to A. Here are key matrices in this chapter:

Coefficient matrix $\boldsymbol{A}$	Upper triangular $\boldsymbol{U}$	Lower triangular $\boldsymbol{L}$
Elimination matrix $\boldsymbol{E_{ij}}$	Overall elimination $\boldsymbol{E}$	Inverse matrix $\boldsymbol{A^{-1}}$
Permutation matrix $\boldsymbol{P}$	Transpose matrix $\boldsymbol{A^{\mathrm{T}}}$	Symmetric matrix $\boldsymbol{S = S^{\mathrm{T}}}$

Our goal is to explain all the steps from $\boldsymbol{A}$ to $\boldsymbol{EA = U}$ to $\boldsymbol{A = E^{-1}U = LU}$ to $\boldsymbol{x}$. (If the steps fail, this signals that $A\boldsymbol{x} = \boldsymbol{b}$ has no solution for most $\boldsymbol{b}$.) Every computer system has a code to find the triangular U and then the solution $\boldsymbol{x}$. Those codes are used so often that elimination adds up to the greatest cost in all of scientific computing.

We need to assume that A has independent columns



$Ax = b \rightarrow Ux = c$

Elimination and Back Substitution

1. Elimination subtracts ℓ_{ij} times row j from row i, leave a zero in row i.
2. $A\boldsymbol{x} = \boldsymbol{b}$ becomes $U\boldsymbol{x} = \boldsymbol{c}$ (or else $A\boldsymbol{x} = \boldsymbol{b}$ is proved to have no solution).
3. Then $U\boldsymbol{x} = \boldsymbol{c}$ is solved by **back substitution** because U is upper triangular.

Elimination and Back Substitution

- There may be
  - no vector that solves $Ax = b$,
  - there may be exactly one solution,
  - there may be infinitely many solution vectors
- Our job it to find all solutions

Success

Elimination

Failure

1 Exactly one solution to $A\boldsymbol{x} = \boldsymbol{b}$. In this case A has independent columns. The rank of A is 2. The only solution to $A\boldsymbol{x} = \mathbf{0}$ is $\boldsymbol{x} = 0$. **A has an *inverse matrix* A^{-1}.**

Example with one solution $(x, y) = (1, 1)$
Independent columns $(2, 4)$ and $(3, 2)$

$$\begin{aligned} 2x + 3y &= 5 \\ 4x + 2y &= 6 \end{aligned} \qquad \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{2} \end{bmatrix}$$

2 No solution to $A\boldsymbol{x} = \boldsymbol{b}$. In this case $\boldsymbol{b}$ is not a combination of the columns of A. In other words $\boldsymbol{b}$ is not in the column space of A. The rank of A is 1.

Example with no solution
Dependent columns $(2, 4)$ and $(3, 6)$

$$\begin{aligned} 2x + 3y &= 6 \\ 4x + 6y &= 15 \end{aligned} \qquad \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{6} \end{bmatrix}$$

Subtract 2 times the first equation from the second to get $0 = 3$. **No solution.**

3 There will be **infinitely many solutions** to $AX = 0$ when the columns of A are not independent. This is the meaning of **dependent columns**—many ways to produce the zero vector $\boldsymbol{b} = \mathbf{0}$. We can multiply $\boldsymbol{X}$ by any number α.

If there is one solution to $A\boldsymbol{x} = \boldsymbol{b}$ then we can add any solution to $AX = 0$. All the vectors $\boldsymbol{x} + \alpha X$ solve the same equations, so we have many solutions.

For any number α $\qquad A(\boldsymbol{x}+\alpha\boldsymbol{X}) = A\boldsymbol{x}+\alpha A\boldsymbol{X} = \boldsymbol{b}+\mathbf{0} = \boldsymbol{b}. \qquad (1)$

Example with infinitely many solutions
***A* has dependent columns: *b* is in C(*A*)**

$$\begin{aligned} 2x + 3y &= 6 \\ 4x + 6y &= 12 \end{aligned} \qquad \begin{bmatrix} \mathbf{2} & \mathbf{3} \\ \mathbf{4} & \mathbf{6} \end{bmatrix}$$

Upper Triangular Matrix

$$\boldsymbol{Ux} = \boldsymbol{c} \quad \text{is} \quad \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix}$$

- 2, 5, 7 are pivots (on main diagonal) and should not equals to 0

Questions

(1)

$$\begin{aligned} 2x &+ 4y - 2z = 2 \\ 4x &+ 9y - 3z = 8 \\ -2x &- 3y + 7z = 10 \end{aligned}$$

(2)

$$\begin{aligned} 2x &- 3y \phantom{{}+3z} = 3 \\ 4x &- 5y + z = 7\,. \\ 2x &- y - 3z = 5 \end{aligned}$$

Matrix Operations (A to U and b to c)

First comes a matrix A (independent columns) that will require no row exchanges. We will apply elimination matrices E_{21} then E_{31} then E_{32}. A and $\boldsymbol{b}$ will change to U and $\boldsymbol{c}$.

The starting matrix is A
The first pivot is 2
The right side is $\boldsymbol{b}$

$$A = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ 4 & 11 & 14 \\ 2 & 8 & 17 \end{bmatrix} \qquad \boldsymbol{b} = \begin{bmatrix} 19 \\ 55 \\ 50 \end{bmatrix} \tag{2}$$

E_{21} multiplies equation 1 by 2 and subtracts from equation 2. You see the new zero.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \qquad \boldsymbol{E_{21}A} = \begin{bmatrix} \mathbf{2} & 3 & 4 \\ \mathbf{0} & 5 & 6 \\ 2 & 8 & 17 \end{bmatrix} \qquad \boldsymbol{E_{21}b} = \begin{bmatrix} 19 \\ \mathbf{17} \\ 50 \end{bmatrix} \tag{3}$$

Matrix Operations (A to U and b to c)

$$E_{31} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \qquad E_{31}E_{21}A = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 5 & 13 \end{bmatrix} \qquad \begin{bmatrix} 19 \\ 17 \\ 31 \end{bmatrix} \tag{4}$$

Move now to column 2 and row 2 (the second pivot row). The pivot is 5, on the diagonal. To eliminate the 5 below it, multiply row 2 by the number 1 and subtract from row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \qquad U = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 6 \\ 0 & 0 & 7 \end{bmatrix} \qquad c = \begin{bmatrix} 19 \\ 17 \\ 14 \end{bmatrix} \tag{5}$$

$E_{32}E_{31}E_{21}A = U$ **is triangular.** $x = (4, 1, 2)$ solved $Ux = c$ on page 41 and $x = (4, 1, 2)$ solves $Ax = b$ here. Since U has 2, 5, 7 on its diagonal we know that back substitution will succeed. The columns of U are independent (and therefore the columns of the original A were independent, as we will see). The matrices A and U have full rank.

Matrix Operations (A to U and b to c)

We can summarize the elimination steps when no row exchanges are involved.

> Use the first equation to produce zeros in column 1 below the first pivot.
>
> Use the new second equation to clear out column 2 below pivot 2 in row 2.
>
> *Continue to column* 3. The expected result is an upper triangular matrix U.

Possible breakdown of Elimination

Elimination might fail. *Zero can appear in a pivot position.* Subtracting that zero from lower rows will not clear out the column below the unwanted zero. Here is an example:

Zero in pivot 2 from elimination in column 1

$$A = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 8 & 17 \end{bmatrix} \to \begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 6 \\ 0 & 5 & 13 \end{bmatrix} = B$$

The cure is simple if it works. **Exchange row 2 with the zero for row 3 with the 5.** Then the second pivot is 5 and we can clear out the second column below that pivot. Elimination continues to U as normal after the row exchange by the matrix P.

Row exchange Successful

$$PB = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 6 \\ 0 & 5 & 13 \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 0 & 5 & 13 \\ 0 & 0 & 6 \end{bmatrix}$$

For this small example, the row exchange is all we need. It produced U with nonzero pivots 2, 5, 6. Normally there are more columns and rows to work on, before we reach U.

Possible breakdown of Elimination

Caution! That row exchange was a success. This is what we hope for, to reach U with no zeros on its main diagonal. (The pivots 2, 5, 6 are on the diagonal.) But a slightly different matrix A^* would lead to a bad situation: **no pivot is available in column 2.**

Dependent columns
U^* is not invertible
A^* is not invertible

$$\rightarrow \quad A^* = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 14 \\ 2 & 3 & 17 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 3 & 4 \\ 0 & 0 & 6 \\ 0 & 0 & 13 \end{bmatrix} = U^* \quad (6)$$

At this point elimination is helpless in column 2. *No second pivot.* This misfortune tells us that **the matrix A^* did not have full rank.** Column 2 of U^* is in the same direction as column 1 of U^*. Column 2 of A^* is in the same direction as column 1 of A^*.

You see how dependent columns are systematically identified by elimination. They can't escape a zero in the pivot. Then there will be nonzero solutions X to $A^*X = \mathbf{0}$. The columns of U^* (and A^*) are not independent.

Elimination and Permutation

There is a simple way to make sure that operations on the matrix A (left side of equations) are also executed on $\boldsymbol{b}$ (right side of equations). The good way is to **include $\boldsymbol{b}$ as an extra column with A**. The combination $\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix}$ is called an **augmented matrix**.

$$\begin{bmatrix} A & \boldsymbol{b} \end{bmatrix} = \begin{bmatrix} 2 & 4 & -2 & \mathbf{2} \\ 4 & 9 & -3 & \mathbf{8} \\ -2 & -3 & 7 & \mathbf{10} \end{bmatrix} \xrightarrow{E} \begin{bmatrix} 2 & 4 & -2 & \mathbf{2} \\ 0 & 1 & 1 & \mathbf{4} \\ 0 & 0 & 4 & \mathbf{8} \end{bmatrix} = \begin{bmatrix} U & \boldsymbol{c} \end{bmatrix}. \qquad (7)$$

Now we include an example that requires a permutation matrix P. It will exchange equations and avoid zero in the pivot. The new matrix A needs P to improve column 2.

Exchange rows 2 and 3

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix} \xrightarrow{E} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 4 & 5 \end{bmatrix} \xrightarrow{P} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & 1 \end{bmatrix} = U$$

Those elimination steps E_{21} and E_{31} and E_{32} produced zeros in positions (2, 1) and (3, 1) and (3, 2). The matrices E have -2 and $+1$ and -1 in those positions. The same steps E_{21}, E_{31}, E_{32} must be applied to the right hand side $\boldsymbol{b}$, to keep the equations correct.

$$\boldsymbol{b} = \begin{bmatrix} 2 \\ 8 \\ 10 \end{bmatrix} \to E_{21}\boldsymbol{b} = \begin{bmatrix} 2 \\ 4 \\ 10 \end{bmatrix} \to E_{31}E_{21}\boldsymbol{b} = \begin{bmatrix} 2 \\ 4 \\ 12 \end{bmatrix} \to E_{32}E_{31}E_{21}\boldsymbol{b} = E\boldsymbol{b} = \boldsymbol{c} = \begin{bmatrix} 2 \\ 4 \\ 8 \end{bmatrix}$$

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$$A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 3 \\ 0 & 4 & 5 \end{bmatrix} \xrightarrow{E} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 4 & 5 \end{bmatrix} \xrightarrow{P} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 4 & 5 \\ 0 & 0 & 1 \end{bmatrix} = U$$

That permutation P_{23} exchanged rows 2 and 3 when it was needed to avoid a zero pivot. But we could have exchanged rows 2 and 3 at the start. Then E_{21} and E_{31} change places.

Question - 2

Gaussian Elimination
vs
Gauss-Jordan elimination

x, y and z?

$$
\begin{aligned}
x + 3y + 2z &= 13 \\
4x + 4y - 3z &= 3 \\
5x + y + 2z &= 13
\end{aligned}
\quad \Rightarrow \quad
\begin{aligned}
x + 3y + 2z &= 13 \\
-8y - 11z &= -49 \\
45z/4 &= 135/4
\end{aligned}
$$

Upper Triangular Matrix

Reduced Row Echelon Form

- Converting augmented matrix into a reduced row echelon form is called Gauss–Jordan elimination.

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & * \\ 0 & 1 & 0 & * \\ 0 & 0 & 1 & * \end{array}\right)$$

- We can read off the x, y and z values directly from this augmented matrix

Inverse Matrix

Suppose A is a square matrix. We look for an "***inverse matrix***" A^{-1} of the same size, so that $\boldsymbol{A^{-1}}$ **times** $\boldsymbol{A}$ **equals** $\boldsymbol{I}$. Whatever A does, A^{-1} undoes. Their product is the identity matrix—which does nothing to a vector, so $A^{-1}A\boldsymbol{x} = \boldsymbol{x}$. *But* A^{-1} *might not exist.*

The n by n matrix A needs n independent columns to be invertible. Then $A^{-1}A = I$.

What a matrix mostly does is to multiply a vector. Multiplying $A\boldsymbol{x} = \boldsymbol{b}$ by A^{-1} gives $A^{-1}A\boldsymbol{x} = A^{-1}\boldsymbol{b}$. ***This is*** $\boldsymbol{x} = A^{-1}\boldsymbol{b}$. The product $A^{-1}A$ is like multiplying by a number and then dividing by that number. Numbers have inverses if they are not zero. Matrices are more complicated and interesting. The matrix A^{-1} is called "$\boldsymbol{A}$ **inverse**".

> **DEFINITION** The matrix A is ***invertible*** if there exists a matrix A^{-1} that "inverts" A:
>
> **Two-sided inverse** $\qquad \boldsymbol{A^{-1}A = I}$ and $\boldsymbol{AA^{-1} = I}$. $\qquad$ (4)

The Facts About Inverse Matrices

- Not all matrices have inverses. Columns must be independent.
- Notes:
  - The inverse exists if and only if elimination produces n pivots (row exchanges are allowed). Elimination solves $Ax = b$ without explicitly using the matrix A^{-1}
  - The matrix A cannot have two different inverses.
  - If A is invertible, the one and only solution to $Ax = b$ is $x = A^{-1}b$:

$$\boxed{\textit{Multiply} \quad Ax = b \quad \textit{by} \quad A^{-1}. \quad \textit{Then} \quad x = A^{-1}Ax = A^{-1}b.}$$

The Facts About Inverse Matrices

- Not all matrices have inverses. Columns must be independent.
- Notes:
  - The inverse exists if and only if elimination produces n pivots (row exchanges are allowed). Elimination solves $Ax = b$ without explicitly using the matrix A^{-1}
  - The matrix A cannot have two different inverses.
  - If A is invertible, the one and only solution to $Ax = b$ is $x = A^{-1}b$
  - A square matrix is invertible if and only if its columns are independent.
  - A 2 by 2 matrix is invertible if and only if the number ad — bc is not zero

2 by 2 Inverse $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Example 3 Three of these matrices are invertible, and three are singular. Find the inverse when it exists. Give reasons for noninvertibility (zero determinant, too few pivots, nonzero solution to $A\boldsymbol{x} = \mathbf{0}$) for the other three. The matrices are in the order $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}, \boldsymbol{D}, \boldsymbol{S}, \boldsymbol{T}$:

$$\begin{bmatrix} 4 & 3 \\ 8 & 6 \end{bmatrix} \quad \begin{bmatrix} 4 & 3 \\ 8 & 7 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 0 \end{bmatrix} \quad \begin{bmatrix} 6 & 6 \\ 6 & 6 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

The Inverse of a Product AB

If A and B are invertible (same size) then the inverse of AB is $B^{-1}A^{-1}$.

$$(AB)^{-1} = B^{-1}A^{-1} \qquad (AB)(B^{-1}A^{-1}) = AIA^{-1} = AA^{-1} = I \quad (7)$$

$B^{-1}A^{-1}$ illustrates a basic rule of mathematics: Inverses come in reverse order. It is also common sense: If you put on socks and then shoes, the first to be taken off are the _____. The same reverse order applies to three or more matrices:

$$\textbf{Reverse order} \qquad (ABC)^{-1} = C^{-1}B^{-1}A^{-1} \qquad (8)$$

Example 4 ***Inverse of an elimination matrix.*** If E subtracts 5 times row 1 from row 2, then E^{-1} *adds* 5 times row 1 to row 2:

E subtracts
E^{-1} adds

$$E = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{-5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{and} \quad E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ \mathbf{5} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Multiply EE^{-1} to get the identity matrix I. Also multiply $E^{-1}E$ to get I. We are adding and subtracting the same 5 times row 1. If $AC = I$ then for square matrices $CA = I$.

Example 5 Suppose F subtracts 4 times row 2 from row 3, and F^{-1} adds it back:

$$F = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 1 \end{bmatrix} \quad \text{and} \quad F^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 4 & 1 \end{bmatrix}.$$

Non-invertible (singular) matrices

- If we cannot convert *(**A**| **I**)* into the augmented matrix (I | $\mathbf{A}^{-1}$) then matrix **A** is non-invertible (singular).

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 3 & 5 \\ 2 & 5 & 6 & 9 \\ -3 & 1 & 2 & 3 \\ 1 & 13 & -30 & -49 \end{pmatrix}$$

Non-invertible (singular) matrices

$$\begin{array}{c} R_1 \\ R_2 \\ R_3 \\ R_4 \end{array} \left(\begin{array}{rrrr|rrrr} 1 & -2 & 3 & 5 & 1 & 0 & 0 & 0 \\ 2 & 5 & 6 & 9 & 0 & 1 & 0 & 0 \\ -3 & 1 & 2 & 3 & 0 & 0 & 1 & 0 \\ 1 & 13 & -30 & -49 & 0 & 0 & 0 & 1 \end{array}\right)$$

$$\left(\begin{array}{rrrr|rrrr} 1 & -2 & 3 & 5 & 1 & 0 & 0 & 0 \\ 0 & 9 & 0 & -1 & -2 & 1 & 0 & 0 \\ 0 & -5 & 11 & 18 & 3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 8 & 0 & 3 & 1 \end{array}\right)$$

$rref\,(\mathbf{A}) = \mathrm{R}$ has at least one row of zeros $\Leftrightarrow$ $\mathbf{A}$ is non-invertible.

Question

What three elimination matrices E_{21}, E_{31}, E_{32} put A into its upper triangular form $E_{32}E_{31}E_{21}A = U$? Multiply by E_{32}^{-1}, E_{31}^{-1} and E_{21}^{-1} to factor A into L times U:

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix} \quad \text{and} \quad L = E_{21}^{-1}E_{31}^{-1}E_{32}^{-1}.$$